

Voluntary wheel running behavior is reduced by SARS-CoV-2 S1 administration and mitigates hypothalamic neuroinflammatory processes

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ABSTRACT

Post-acute sequelae of SARS-CoV-2 infection (PASC) is a major public health consequence of COVID-19. Persistent PASC symptoms affect multiple physiological systems, including the nervous system. The etiology of PASC is unknown; however, brain-mediated symptoms may arise from dysregulated neuroinflammatory processes. We have reported that intra-cisterna magna (ICM) administration of the SARS-CoV-2 spike (S)1 antigen in rats induces behavioral alterations, prolonged neuroinflammation, and reduced brain corticosterone. Inappropriate neuroinflammation triggered by immune challenge suppresses motivated behaviors, including voluntary wheel running (VWR), and disrupts diurnal rhythms. Accordingly, this study tested the hypothesis that S1 reduces VWR distance and changes diurnal activity patterns. Given evidence that regular physical activity reduces inflammation, we also determined whether VWR attenuates the neuroinflammatory effects of S1. Male Sprague-Dawley rats were housed with either locked (sedentary) or unlocked (VWR) running wheels. After 12 days, rats received ICM S1 or vehicle and remained undisturbed while VWR was recorded for 17 days. S1-treated rats ran less and showed disrupted diurnal VWR patterns compared with vehicle controls. Eighteen days post-ICM injections, sedentary S1-treated rats, but not VWR rats, exhibited persistent elevation of hypothalamic gene expression of IL1 β , MhcII, Tlr2, and Tlr4. In contrast, VWR S1-treated rats, but not sedentary S1-treated rats, had elevated hypothalamic corticosterone 18 days post-injection. Taken together, S1 in the brain produces behavioral symptoms of PASC and prolonged neuroinflammation. Despite reductions in total running distance after S1, VWR was sufficient to block the persistent upregulation of neuroinflammatory genes. Brain corticosterone may contribute to the impacts of S1 and VWR.

1. Introduction

Long COVID, or post-acute sequelae of SARS-CoV-2 (PASC), is a persisting, remitting syndrome characterized by a constellation of symptoms that endure beyond the acute phase of SARS-CoV-2 infection (Al-Aly et al., 2024). In approximately 10% of patients (Davis et al., 2023), a diagnosis of PASC is made after partial or complete recovery from an acute infection, with symptoms persisting for at least 3 months (Huerne et al., 2023). Symptoms of PASC are heterogeneous and affect multiple physiological systems, including the nervous system.

Brain-related PASC symptoms span cognitive and behavioral domains. Cognitive symptoms, including deficits in attention and working memory, have been shown to be associated with reduced frontal-parietal and frontal-occipital connectivity (Niemczak et al., 2025; Ran et al., 2026). Behavioral symptoms, the focus of this study, include exercise intolerance, post-exertional malaise, fatigue, and diurnal activity disruptions (An et al., 2024; Das et al., 2024; Gil et al., 2023a; Kolenicova et al., 2023; Stussman et al., 2025; Wilson et al., 2025). In general, patients with PASC experience a decline in physical activity, which may

contribute to additional health problems, including reduced cardiopulmonary fitness (Vogel et al., 2026).

Exercise intolerance and its more severe form, post-exertional malaise (PEM), are characterized by extreme fatigue after physical activity that was previously nonchallenging. Both exercise intolerance and PEM are aversive and likely contribute to the notable decrease in physical activity among patients with PASC (Aschman et al., 2023; Durstenfeld et al., 2023; Zheng et al., 2024). Gil et al. report that there is a 57% higher chance of physical inactivity among patients with PASC (Gil et al., 2023b). Furthermore, Vélez-Santamaría et al. found a significant association between PASC and lower physical activity, with 44.7% of patients reporting a decline in physical activity since their PASC diagnosis (Vélez-Santamaría et al., 2023).

PASC patients also report daytime sleepiness and altered wake/sleep cycles (Batool-Anwar et al., 2025; Coelho et al., 2024; Wilson et al., 2025). Both reduced physical activity (Say et al., 2025; Turner and Reed, 2022) and disrupted diurnal activity patterns (Almeneessier and BaHammam, 2018; Batool-Anwar et al., 2025; Nishimon et al., 2019) contribute to sleep disturbances and generalized fatigue

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(Clemente-Suarez et al., 2026; Huang et al., 2026). The neural and peripheral mechanisms underlying these behavioral symptoms remain unknown.

Findings from animal studies and clinical evidence reveal a causal link between neuroinflammation and brain-mediated outcomes (Fontes-Dantas et al., 2023; Frank et al., 2022, 2023b, 2024). Evidence indicates that SARS-CoV-2 antigens can enter the brain, trigger inappropriate inflammation, and affect behavior. Rhea et al. demonstrated that SARS-CoV-2 antigens injected into the bloodstream can cross the blood-brain barrier and infiltrate brain tissue (Rhea et al., 2021). Erickson et al. determined that S1 directly affects endothelial cells and reduces blood-brain barrier integrity (Erickson et al., 2021). Frank et al. reported that intra-cisterna magna (ICM) injection of S1 into male rats increases microglial neuroinflammatory gene expression and protein levels in a TLR2-and TLR4-dependent manner (Frank et al., 2022, 2023a, 2024). Furthermore, ICM injection of S1 produced anxiety-like behavior, disrupted sleep, and altered naturalistic home-cage behaviors in male rats (Frank et al., 2022, 2024). Finally, SARS-CoV-2 antigen also reduces serum and brain corticosterone, potentially sustaining neuroinflammatory processes (Frank et al., 2024).

Clinical and preclinical studies suggest that regular physical activity reduces neuroinflammation (Alves et al., 2025; Feng et al., 2026; Kinoshita et al., 2026; Kruijff et al., 2026; Littlefield et al., 2015; Sahoo et al., 2026; Speaker et al., 2014; Zhou et al., 2026). VWR is sufficient to reduce inappropriate neuroinflammation following exposure to neuroinflammatory triggers, including lipopolysaccharide (LPS), stress, and aging (Alves et al., 2025; Grabowski et al., 2025; Grace et al., 2016; Littlefield et al., 2015; Martin et al., 2013; Speaker et al., 2014). Furthermore, several clinical studies have reported decreased serum glucocorticoid levels (e.g., cortisol) in patients with PASC (Klein et al., 2023; Marques et al., 2023; Porntharukchareon et al., 2024). Hypocortisolism can cause fatigue (Garcia-Gonzalez et al., 2024) and allow inflammation to go unchecked. Taken together, the clinical and preclinical literature suggests that SARS-CoV-2 antigens present in the serum could readily cross the blood-brain barrier and trigger neuroinflammatory processes. Physical inactivity and reduced anti-inflammatory cortisol may contribute to the persistence of inappropriate neural inflammation and PASC brain-mediated symptoms (Aoyagi et al., 2025; Kruijff et al., 2026; Liu et al., 2026; Wang et al., 2026; Zhou et al., 2026).

It is therefore important to establish an animal model that enables future studies to explore the links among physical inactivity, brain cortisol, and sustained neuroinflammation, and to develop interventions that restore these key anti-inflammatory mechanisms. Voluntary wheel running (VWR) is a non-stressful, innately motivated, and rewarding behavior that increases physical activity in rats (Dishman et al., 2006; Grace et al., 2016; Greenwood and Fleshner, 2008, 2019; Greenwood et al., 2009, 2011; Herrera et al., 2016; Kennedy et al., 2005; Sasse et al., 2008; Thompson et al., 2016a). Furthermore, VWR is an ideal method for assessing disruptions in diurnal activity patterns (Bowers et al., 2020; Clark et al., 2014; Greenwood et al., 2013, 2014; Thompson et al., 2013, 2016b).

The current study therefore tested the hypothesis that S1 leads to persistent reductions in VWR behavior and changes in diurnal activity patterns. Given evidence that regular physical activity reduces inappropriate inflammation, we also tested whether VWR attenuates S1's effects on neuroinflammatory processes and corticosterone in the hypothalamus, a region previously shown to be responsive to S1 and to orchestrate the behavioral outcomes assessed in this study.

2. Methods

2.1. Animals

Male Sprague-Dawley rats (60–90 days of age; 200–250 g; Envigo, Indianapolis, IN) were single-housed on a 12-h light-dark cycle (lights

on at 0700h) and housed in a specific-pathogen-free barrier facility. Animal care was provided once per week, and food (standard laboratory chow) and water were available *ad libitum*. Rats were acclimated to vivarium conditions for one week before experimentation. All experiments were approved by the Institutional Animal Care and Use Committee of the University of Colorado Boulder (Protocol Number 03004) in compliance with the National Institutes of Health *Guide for the Care and Use of Laboratory Animals* as well as the ARRIVE guidelines for animal use.

2.2. Experimental Design (see Fig. 1)

On experimental day −7, male Sprague-Dawley rats (N = 30) arrived at our facility and were immediately single-housed with a locked running wheel. After a 1-week acclimation period in the vivarium, rats were randomly assigned to either a sedentary (N = 10) or a voluntary wheel-running (VWR; N = 20) group (experimental day 0). We assigned more animals to the VWR group to account for known variability in wheel-running behavior (Garvey et al., 2015). Sedentary rats remained in home cages with locked running wheels for the remainder of the study. Running wheels were unlocked for VWR rats on experimental day 0, and rats were allowed to run for 12 days to stabilize running patterns. On experimental day 12, all rats were randomly assigned to receive either S1 antigen (S1; 1.0 µg) or saline (Veh; sterile 1x PBS) via intra-cisterna magna injection (N = 5 Veh/Sed; N = 5 S1/Sed; N = 10 Veh/VWR; N = 10 S1/VWR). Rats were then left undisturbed for the remainder of the study. Wheel-running behavior of VWR rats was evaluated through the end of the study. On experimental day 30 (18 days post-treatment), rats were saline-perfused, and the hypothalamus was dissected.

2.3. VWR

All rats were housed individually in standard Nalgene Plexiglas cages (45 × 25.2 × 14.7 cm) with a stainless-steel open running wheel (49.5 × 26.4 × 36 cm) attached. Wheels were locked during the acclimatization period (Exp. Day −7 – Exp. Day 0). For sedentary animals, wheels remained locked throughout the experiment. Following acclimatization (Exp. Day 0), wheels for VWR animals were unlocked. VWR animals were allowed to run *ad libitum* for the remainder of the study. Daily wheel revolutions were recorded using VitalView Automated Data-Acquisition software (Starr Life Sciences, Oakmont, PA, USA). Distance (m) was calculated by multiplying the wheel circumference (1.081 m) by the number of revolutions recorded for each animal.

2.4. SARS-CoV-2 S1 subunit antigen

The full-length S1 subunit of recombinant SARS-CoV-2 was obtained from Raybiotech (cat# 230-30161). The S1 antigen (Val16-Gln690) is produced in HEK293 cells, free of the immunogen lipopolysaccharide, and is sequence-matched to the original Wuhan-Hu-1 isolate (Wu et al., 2020). The injected S1 antigen is identical to that used by Rhea et al., who demonstrated that it crosses the blood-brain barrier into the brain parenchyma after intravenous (IV) injection (Rhea et al., 2021).

2.5. Intra-cisterna magna (ICM) injection

To avoid peripheral inflammatory impacts that could be induced by IV injection and to directly assess effects in the CNS, we injected the S1 subunit (1.0 µg) or vehicle (2.5 µg, 0.2 µm filtered, sterile 1x PBS, pH 7.4) ICM. The S1 subunit dose is based on prior studies demonstrating the neuroinflammatory effects of the S1 subunit (Frank et al., 2022, 2024). This procedure does not produce detectable inflammatory responses, and substances injected ICM reach forebrain regions in the CNS (Frank et al., 2012). Proescholdt and colleagues also reported that ICM injection of substances produces widespread distribution throughout the

rat brain (Proescholdt et al., 2000). Briefly, rats were anesthetized with 5% isoflurane in oxygen and maintained on 3% isoflurane during the procedure (estimated duration: ~3 min). The dorsal region of the skull was shaved and swabbed with 70% EtOH. A sterile 27-gauge needle attached via sterile PE50 tubing to a 25.0 μ l Hamilton syringe was inserted into the cisterna magna (verified by withdrawing 2.0 μ l of clear CSF), and the drug was injected over a 30 s period. After injection, the needle was left in place for 30 s to allow diffusion of the drug.

2.6. Tissue dissection

At the conclusion of the study, animals received a lethal intraperitoneal (IP) dose of sodium pentobarbital (150.0 mg/kg). After full anesthesia, cardiac blood was collected, followed by a 3-min transcardiac perfusion with cold saline (0.9%) to remove peripheral immune leukocytes from the CNS vasculature. Brains were rapidly extracted, and the hypothalamus was bisected for protein and mRNA analyses. Tissues were immediately flash-frozen in liquid nitrogen and stored at -80°C .

2.7. Tissue processing for protein assay

Hypothalamic samples were sonicated on ice in a tissue extraction reagent (Invitrogen, cat#: FNN0071) supplemented with a protease inhibitor cocktail (Sigma-Aldrich, cat#: P2714). Homogenates were centrifuged at $14,000 \times g$ for 10 min at 4°C , and supernatants were collected and stored at -80°C . Total protein was quantified using a Bradford assay.

2.8. Enzyme-linked immunosorbent assay (ELISA)

An ELISA for rat corticosterone (Enzo, cat#: ADI-900-097) was performed on hypothalamic protein lysates. The ELISA was run according to the manufacturer's instructions, and protein and steroid levels were normalized to total protein.

2.9. Real-time RT-PCR measurement of gene expression

Total RNA was isolated from the hypothalamus using TRI Reagent (Millipore Sigma, cat#: 93289) and a standard phenol:chloroform extraction (Chomczynski and Sacchi, 1987). Total RNA was quantified using a NanoDrop 2000 spectrophotometer (ThermoFisher). cDNA synthesis was performed using the SuperScript II Reverse Transcriptase kit (ThermoFisher, cat#: 18064014). A detailed description of the PCR amplification protocol has been published previously (Frank et al., 2006). cDNA sequences were obtained from GenBank at the National Center for Biotechnology Information (NCBI; <https://www.ncbi.nlm.nih.gov>). Primer sequences are provided in Supplemental Table 1. Primer specificity was tested using the Basic Local Alignment Search Tool at NCBI (Altschul et al., 1997). Primers were obtained from ThermoFisher. Melt curve analyses verified primer specificity. All primers were designed to span exon-exon boundaries and thus exclude amplification of genomic DNA. PCR amplification of cDNA was performed using the Quantitect SYBR Green PCR Kit (Qiagen, cat#: 204145). The formation of the PCR product was monitored in real time using the CFX96 Touch Real-Time PCR Detection System (Bio-Rad). Relative gene expression was determined using *Actb* as the housekeeping gene and the $2^{-\Delta\Delta\text{CT}}$ method (Livak and Schmittgen, 2001). Neuroinflammatory state was assessed by measuring the following: proinflammatory cytokine (*IL1 β*), microglia/brain macrophage activation markers (*MhcII*), inflammasome (*Nlrp3*), TLR2 (*Tlr2*), and TLR4 (*Tlr4*). The genes were chosen based on prior evidence showing the largest and most enduring S1-induced changes in the hypothalamus (Frank et al., 2022).

2.10. Statistical analysis

Running distance analyses are presented as mean $\pm$ SEM and were

analyzed using a repeated-measures ANOVA with Bonferroni-Holm post hoc tests to correct for multiple comparisons. Where appropriate, running distance was computed as a percentage of baseline running, and data were analyzed using a repeated-measures ANOVA with Bonferroni-Holm post hoc tests to correct for multiple comparisons. Area under the curve (AUC) was computed for daily running distances as a percent of baseline running, and unpaired t-tests were performed between the vehicle and S1 groups. The threshold for statistical significance was set at $\alpha = 0.05$. Sample sizes are provided in figure captions.

ClockLab (Actimetrics), as previously described (Thompson et al., 2021), was used to assess diurnal running-wheel patterns and to determine the acrophase, bathyphase, onset, and offset of running behavior. Acrophase and bathyphase are automatically calculated in the ClockLab analysis by fitting a sine function to the daily activity. Acrophase and bathyphase indicate the peak (maximum) and trough (minimum) of the resulting sine wave. Onset and offset times are estimated using a least-squares fit algorithm on the daily activity data. For each marker, the data are initially exported at the circadian hour of the event based on a 24-h period. Diurnal phase markers, acrophase (peak), bathyphase (trough), onset, and offset of activity were assessed as a percent change from baseline. Area under the curve (AUC) was computed for the percent of baseline for all diurnal markers and unpaired t-tests were performed between the vehicle and S1 groups.

All neuroimmune measures are presented as % of vehicle control and were analyzed using a two-way ANOVA with unprotected Fisher's LSD post hoc tests to correct for multiple comparisons. Analyses were conducted using Prism 10 (GraphPad Software, LLC).

3. Results

3.1. SARS-CoV-2 S1 subunit treatment reduces VWR distance

The average total running distance (m) across the study was lower in S1- than in vehicle-injected animals. The reduction in running distance emerged 7 days after injection (Supplemental Fig. 1). When analyzed as a percent of baseline running, VWR was reduced 1 week after S1 compared with vehicle and remained lower throughout the study ($F_{(1,18)} = 6.73$, $p = 0.018$; Fig. 2A). Area under the curve analysis revealed a significant difference in total distance run between groups ($df = 16$, $t = 3.738$, $p = 0.0018$; Fig. 2B).

3.2. SARS-CoV-2 S1 subunit alters the diurnal pattern of VWR

3.2.1. Acrophase and bathyphase

Compared with vehicle-treated rats, S1 induced a misalignment in the acrophase or peak of running behavior (Fig. 3A). Following S1 administration, S1 animals displayed, on average, a 6.13% shift in the time of peak activity, from 21:52 (9:52 p.m.) to 22:49 (10:49 p.m.). In contrast, vehicle animals displayed, on average, a 1.92% shift in peak activity from 21:52 (9:52 p.m.) to 21:55 (9:55 p.m.). AUC was computed for vehicle and S1 treatment groups to provide a summary measure across time (17 days) post-injection ($df = 18$, $t = 2.335$, $p = 0.0313$; Fig. 3B).

Prior S1 treatment also produced a misalignment in the bathyphase, the lowest level or trough of running behavior (Fig. 3C). Following S1 administration, S1 animals displayed, on average, a 15.27% shift in the activity trough from 09:51 (9:51 a.m.) to 10:58 (10:58 a.m.). In contrast, vehicle animals displayed, on average, a 5.42% shift in the activity trough from 09:51 (9:51 a.m.) to 10:01 (10:01 a.m.). AUC was computed for vehicle and S1 treatment groups to provide a summary measure across time (17 days) post-injection ($df = 18$, $t = 2.278$, $p = 0.0351$; Fig. 3D). S1-injected animals exhibited a shift toward a later acrophase and bathyphase, demonstrating a significant effect of S1 injection on diurnal running patterns.

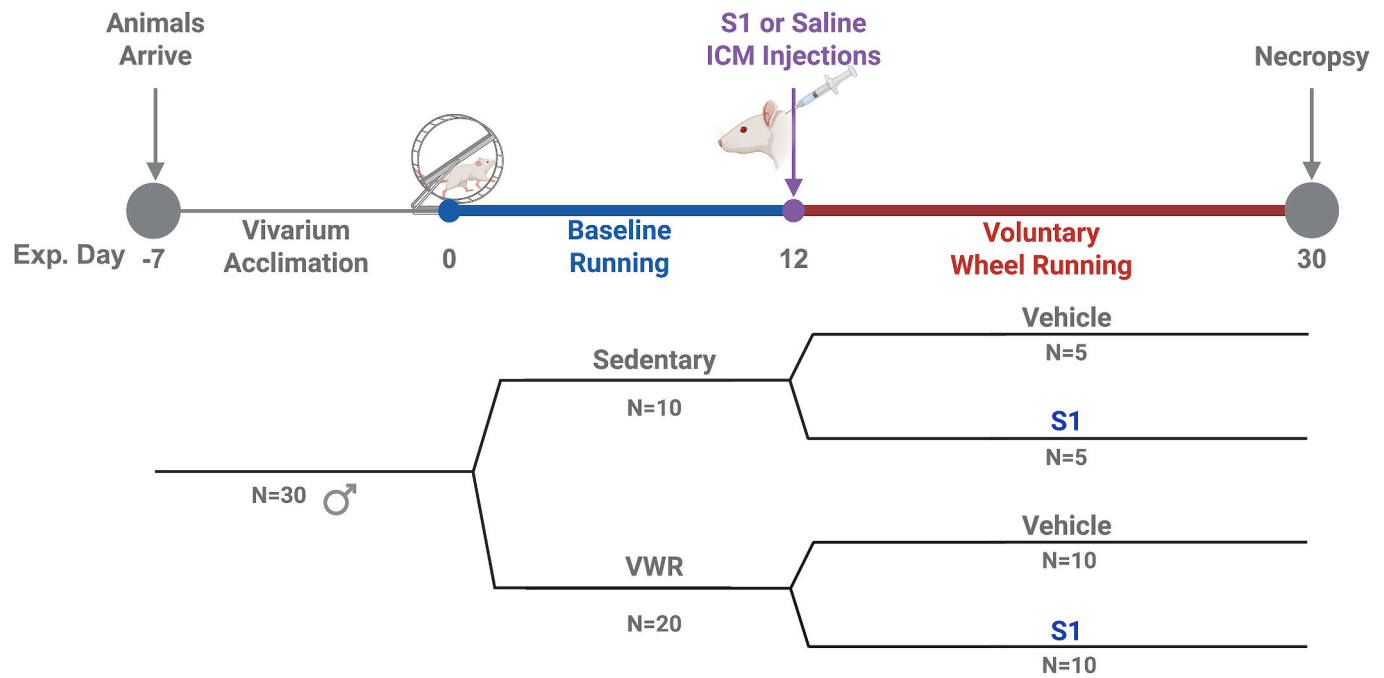


Fig. 1. Schematic of Experimental Design. Visualization of experimental days and groups across the experiment. The acclimatization period occurred from experimental day -7 to 0. Wheels were unlocked for VWR rats on experimental day 0. Baseline running was recorded from experimental days 0 to 12. On experimental day 12, all animals were treated with either S1 subunit (1.0 μ g) or vehicle (2.5 μ g, 0.2 μ m filtered, sterile 1x PBS, pH 7.4). Running behavior was then recorded for 18 days post-treatment, until the end of the study (experimental day 30). Figure was prepared using BioRender.

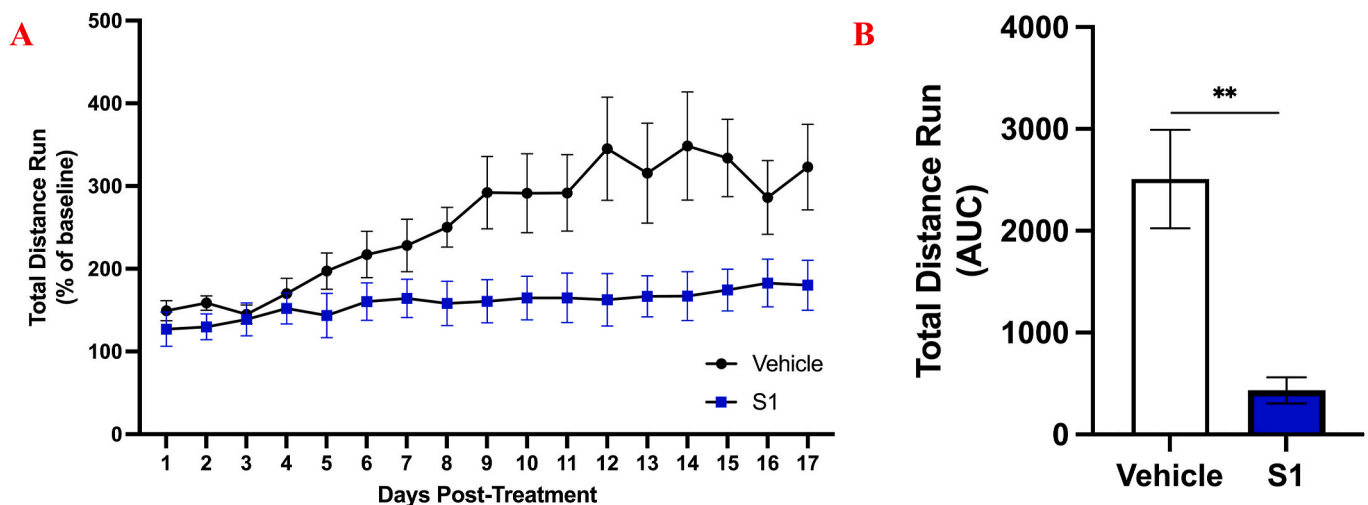


Fig. 2. Effect of Prior S1 Exposure on VWR Patterns. Rats were injected ICM with vehicle (0.9% saline; N = 10) or S1 (1.0 μ g; N = 10). (A) Total running distance was corrected as a % of baseline running, computed over five days before injection (Exp. Days 7 to 11) for both the vehicle and S1 groups. (B) Area under the curve (AUC) was performed for running distances as a percent of baseline to provide a summary measure of the effects of vehicle and S1 on running patterns. S1 vs vehicle, ** $p < 0.01$.

3.2.2. Onset and offset

S1 induced a misalignment in the timing of running onset (Fig. 4A). Following S1 administration, S1 animals displayed, on average, a 9.41% shift in activity onset from 17:45 (5:45 p.m.) to 9:05 (7:05 p.m.), whereas vehicle animals displayed, on average, a 2.08% shift in activity onset from 17:45 (5:45 p.m.) to 17:48 (5:58 p.m.). AUC was computed for the vehicle and S1 treatment groups to provide a summary measure across time (17 days) post-injection, demonstrating a significant shift in activity onset in S1-injected animals ($df = 18$, $t = 2.827$, $p = 0.0112$; Fig. 4B). S1 injection did not affect the offset of running behavior (Fig. 4C and D).

3.3. Effect of VWR on the neuroinflammatory response to the SARS-CoV-2 S1 subunit in the hypothalamus

To determine whether hypothalamic neuroinflammatory gene expression was upregulated 18 days after S1 in physically active compared with sedentary rats, we measured proinflammatory cytokine (*IL1 β*), microglia/brain macrophage activation markers (*MhclII*), inflammasome (*Nlrp3*), TLR2 (*Tlr2*), and TLR4 (*Tlr4*).

3.3.1. Neuroinflammatory gene expression

There were statistically significant interactions between injection (S1, vehicle) and activity status (VWR, sedentary) for *IL1 β* ($F_{(1,26)} = 25.9$,

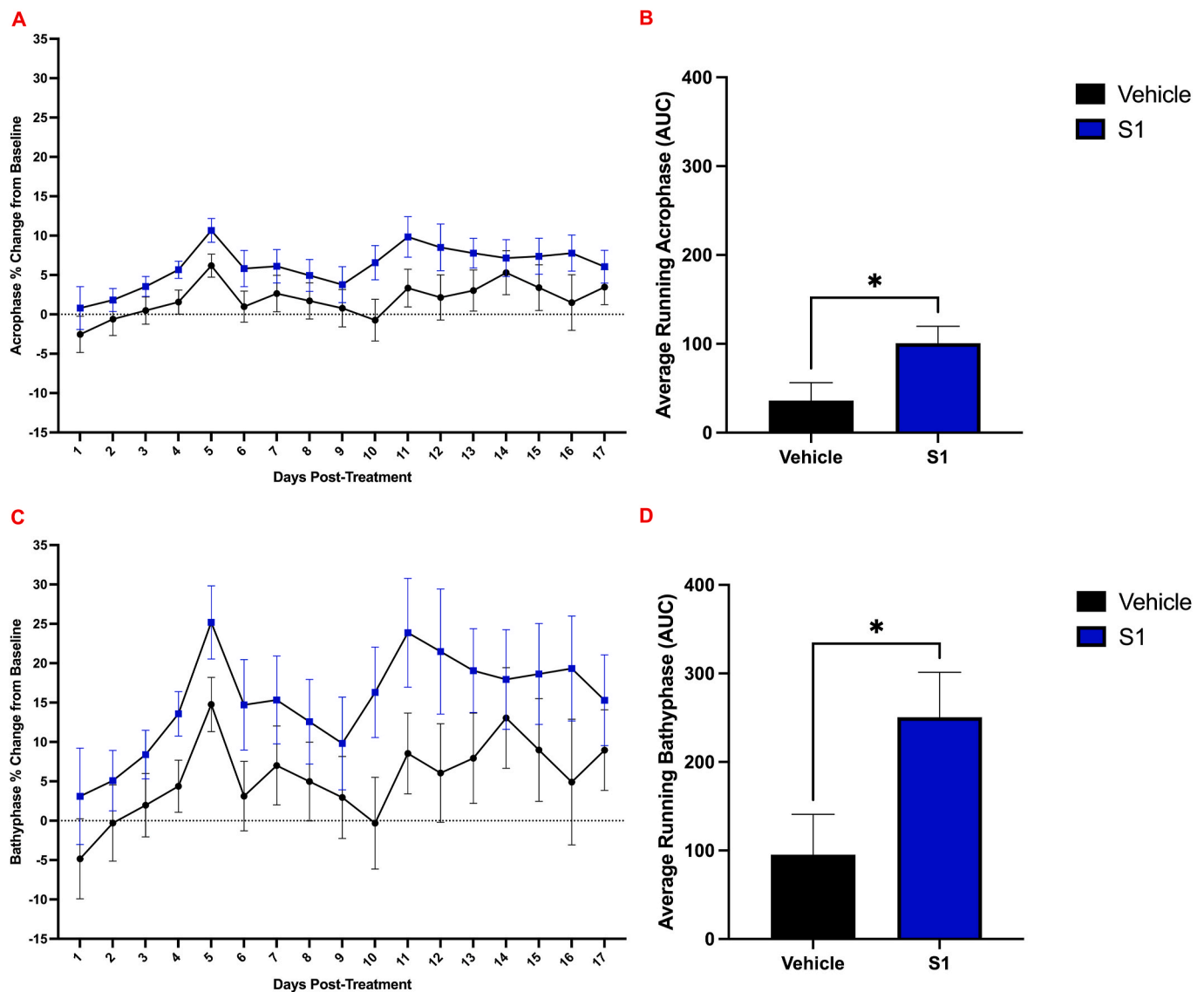


Fig. 3. Effect of Prior S1 Exposure on Acrophase and Bathyphase of Wheel Running Pattern. Rats were injected ICM with vehicle (0.9% saline; $N = 10$) or S1 (1.0 μg ; $N = 10$). Acrophase (peak) of wheel running behavior (A) was measured across all experimental days and was corrected as a % of baseline running, which was computed for five days (Exp. Day 7 to 11) prior to injection, for both vehicle and S1 groups. (B) Area under the curve (AUC) was performed for acrophase as a percent of baseline to provide a summary measure of the effects of vehicle and S1 on running patterns. Bathyphase (trough) of wheel-running behavior (C) was measured across all experimental days and corrected as a % of baseline running, computed over the 5 days (Exp. Day 7 to 11) prior to injection, for both the vehicle and S1 groups. (D) Area under the curve (AUC) was performed for bathyphase as a percent of baseline to provide a summary measure of the effects of vehicle and S1 on running patterns. S1 vs vehicle, * $p < 0.05$.

$p < 0.0001$; Fig. 5A), *MhcII* ($F_{(1,26)} = 15.92$, $p = 0.0005$; Fig. 5B), *Nlrp3* ($F_{(1,26)} = 10.67$, $p = 0.003$; Fig. 5C), *Tlr2* ($F_{(1,26)} = 7.323$, $p = 0.012$; Fig. 6A), and *Tlr4* ($F_{(1,26)} = 22.87$, $p < 0.0001$; Fig. 6B). Post hoc analyses (see figure captions) revealed that 18 days after S1, the % of vehicle was higher in sedentary rats than in VWR rats for every gene except *Nlrp3*.

3.3.2. Corticosterone

There was a statistically significant interaction between injection (S1, vehicle) and activity status (VWR, sedentary) for brain corticosterone ($F_{(1,26)} = 6.794$, $p = 0.015$; Fig. 7). Post hoc analyses (see figure caption) revealed that 18 days after S1, the % of vehicle increased in S1-VWR rats but not in S1-sedentary rats.

4. Discussion

ICM S1 reduces VWR, disrupts diurnal rhythms, and upregulates

markers of microglial priming and activation in the hypothalamus in sedentary rats. Physical activity and brain cortisol regulate neuro-inflammatory processes. Despite reduced total running distance, VWR was sufficient to prevent persistent upregulation of neuroinflammatory gene expression and to increase hypothalamic corticosterone 18 days after S1. The results of this study suggest that S1 induces inappropriate neuroinflammatory processes, which may be sustained by the loss of key anti-inflammatory factors, including physical activity and brain cortisol (Aoyagi et al., 2025; Kruijff et al., 2026; Liu et al., 2026; Wang et al., 2026; Zhou et al., 2026).

4.1. S1 reduces VWR

Our previous work established that S1 is sufficient to produce anxiety-like behavior, disrupted sleep, and changes in naturalistic home-cage behaviors, including reduced total activity, water intake, and self-grooming (Frank et al., 2022). Biological and behavioral symptoms

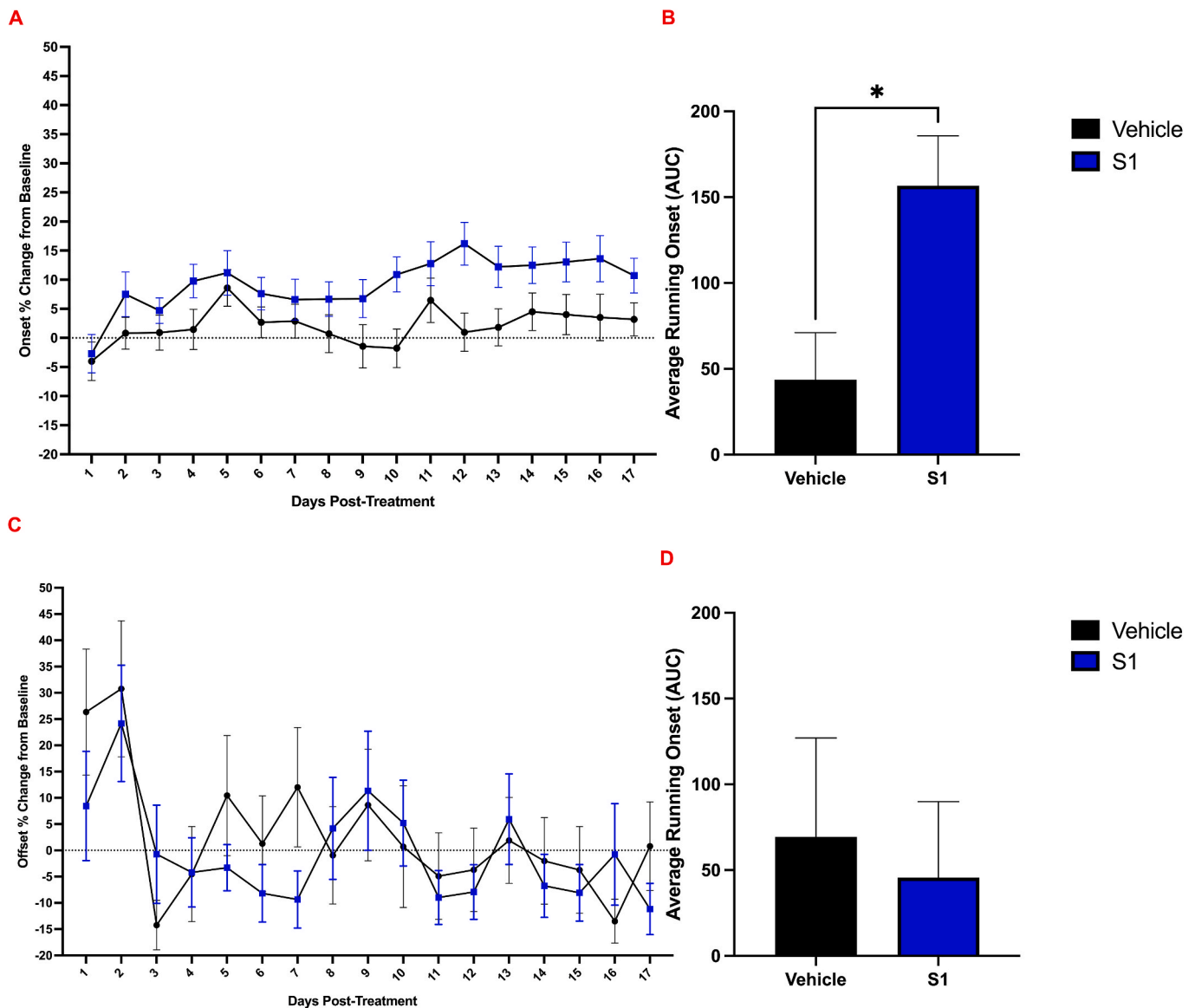


Fig. 4. Effect of Prior S1 Exposure on Onset and Offset of Wheel Running Pattern. Rats were injected ICM with vehicle (0.9% saline; N = 10) or S1 (1.0 μ g; N = 10). Onset of wheel running behavior (A) was measured across all experimental days and was corrected as a % of baseline running, which was computed for five days (Exp. Day 7-11) prior to injection, for both vehicle and S1 groups. (B) Area under the curve (AUC) was performed for onset as a percent of baseline to provide a summary measure of the effects of vehicle and S1 on running patterns. Offset of wheel-running behavior (C) was measured across all experimental days and corrected as a % of baseline running, computed over the five days (Exp. Day 7-11) prior to injection, for both the vehicle and S1 groups. (D) Area under the curve (AUC) was performed for offset as a percent of baseline to provide a summary measure of the effects of vehicle and S1 on running patterns. S1 vs vehicle, * $p < 0.05$.

associated with PASC resemble the acute sickness response; however, it remains unclear whether inappropriate neuroinflammatory states, such as those following S1, contribute to persistent reductions in motivated behavior.

The sickness response is a constellation of adaptive behavioral and physiological responses to infection or injury, common to humans and animals (Dantzer et al., 2008; Saxena and Maze, 2018). A key aspect of the sickness response is the reorganization of motivational states from reward-seeking to coping behaviors after immunological challenges (Dantzer, 2001).

VWR, like human running, is intrinsically motivating and rewarding (Cadejani and Kater, 2017; Heyse et al., 2015; Saul et al., 2017; Say et al., 2025). Rats will bar-press to access a running wheel, further underscoring the intrinsic motivation for VWR (Iversen, 1993). Animal studies show that VWR engages reward-related neurocircuitry and that conditioned place preference can develop for locations associated with a running wheel (Greenwood et al., 2011; Herrera et al., 2016).

Neuroinflammatory processes are well established to play a causal role in the sickness response (Dantzer et al., 2008). Harden and colleagues demonstrated that inflammatory cytokines in the brain contributed to LPS-evoked sickness responses, including fever, decreased body mass, decreased food intake, and VWR (Harden et al., 2011). Reductions in home-cage locomotion and VWR occur within hours and persist for 24-48 h after LPS (Alencar et al., 2026; Harden et al., 2011). In contrast, reduced VWR emerges after 5-7 days and persists for 18 days following ICM S1 injection (Supplemental Fig. 1). This pattern does not align with a typical acute sickness response to a proinflammatory challenge such as LPS (Dantzer et al., 2008), suggesting that S1 triggers atypical sickness responses and that reduced VWR may be more reflective of exercise intolerance than sickness.

4.2. ICM S1 evokes prolonged microglia activation and priming

Inflammatory stimuli, stress, and aging trigger inappropriate

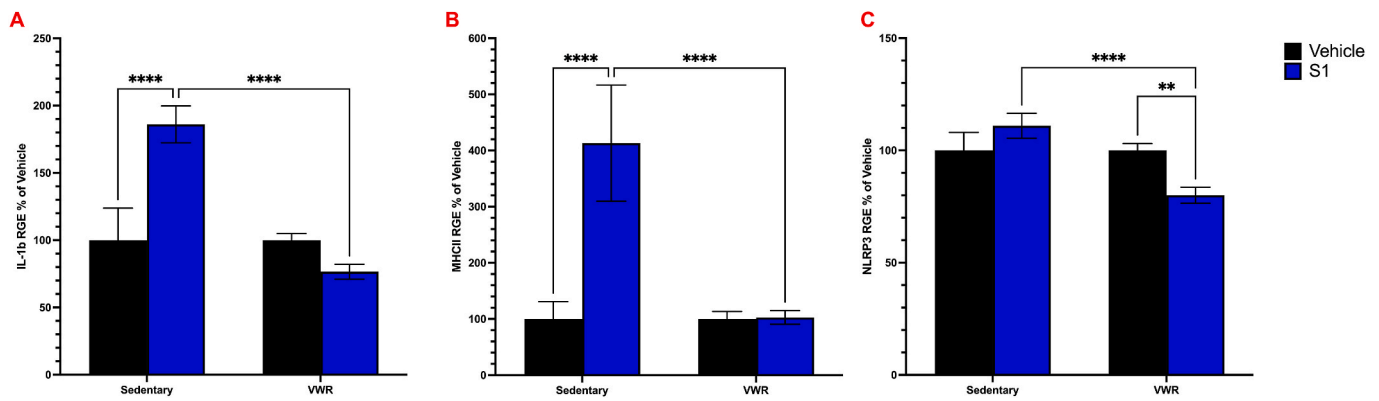


Fig. 5. Effect of VWR on the Neuroinflammatory Effects of S1 (18d post-treatment). Rats were injected ICM with vehicle (0.9% saline) or S1 (1.0 μ g). *IL1 β* (A), *MhcII* (B), and *Nlrp3* (C) were measured in the hypothalamus. Data are reported as a % change from sedentary vehicle animals. N = 5 per sedentary group; N = 10 per VWR group. RGE = relative gene expression. Main effects and post-hoc comparisons are depicted in figures at varying levels of significance, **p < 0.01, ****p < 0.0001.

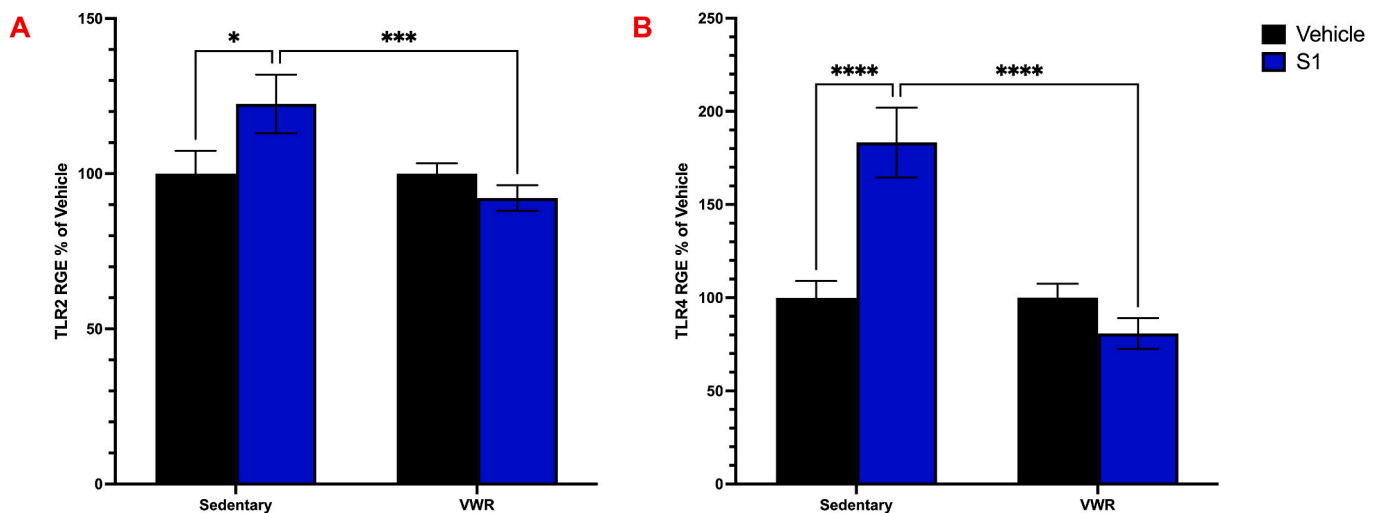


Fig. 6. Effect of VWR on Toll-like Receptors Post-S1-treatment (18d post-treatment). Rats were injected ICM with vehicle (0.9% saline) or S1 (1.0 μ g). *Tlr2* (A) and *Tlr4* (B) were measured in the hypothalamus. Data are reported as a % change from sedentary vehicle animals. N = 5 per sedentary group; N = 10 per VWR group. RGE = relative gene expression. Main effects and post-hoc comparisons are depicted in figures at varying levels of significance, *p < 0.05, ***p < 0.001, ****p < 0.0001.

neuroinflammatory processes (Alves et al., 2025; Frank et al., 2010; Grabowski et al., 2025; Littlefield et al., 2015; Martin et al., 2013; Speaker et al., 2014). Here, we report that 18 days after ICM S1, markers of microglial priming and activation were upregulated in sedentary animals, specifically hypothalamic *IL1 β* , *MhcII*, *Tlr2*, and *Tlr4*. We found no detectable differences in *Nlrp3* gene expression at this time point in sedentary animals.

We previously reported that ICM S1 dose-dependently upregulates genes associated with microglial priming and activation (Frank et al., 2022). Twenty-four hours after rats were injected with S1, microglia/brain macrophage activation markers (*Iba1*, *Cd11b*, *MhcII*), an astrocyte activation marker (*Gfap*), *Tlr4*, *Nlrp3*, *Il1b*, and *Il6* gene expression, and inflammatory protein concentrations (IL-1 β , IL-6, TNF, CXCL1, IL-10) were elevated in the hypothalamus compared with vehicle controls. These findings suggest that S1 acutely induces microglial activation after 8 and 24 h. Furthermore, we demonstrated evidence of persistent microglial priming (Frank et al., 2024). Rats received either ICM S1 or ICM vehicle. Twenty-one days after S1, microglia were isolated and stimulated with LPS. Microglia from rats previously treated with S1, compared with vehicle-treated rats, produced more IL-1 β and IL-6 protein in culture. The current findings extend our previous work and support the conclusion that ICM S1 activates long-lived, inappropriate

neuroinflammatory processes.

4.3. Neuroinflammation and behavior

As previously described, Frank et al. reported evidence that ICM S1 induces persistent microglial priming. S1 priming resulted in greater proinflammatory responses and greater disruptions in home-cage behavior following LPS than in vehicle rats. Behavioral changes included reduced activity, drinking, and self-grooming (Frank et al., 2024). Other groups are revealing potential mechanisms underlying behavioral deficits produced by SARS-CoV-2 antigen in the brain. Yuan et al., using conditional knockout mice, recently showed that depressive-like behaviors after injection of the spike receptor-binding domain into the brain depend on astrocytic Cx43-mediated disruption of intercellular gap junction communication (Yuan et al., 2026). Using whole-cell patch-clamp recordings, Lee et al. demonstrated that inappropriate neuroinflammation and depressive-like behavior following S1 injection into the brain result from upregulation of microglial Kv1.3 channel activity (Lee et al., 2026). Interestingly, fatigue measures obtained using the Rotorad handgrip test were unaffected by S1. Inappropriate neuroinflammation also disrupts diurnal rhythms, including VWR (Bowers et al., 2022; Greenwood and Fleshner, 2019; Thompson

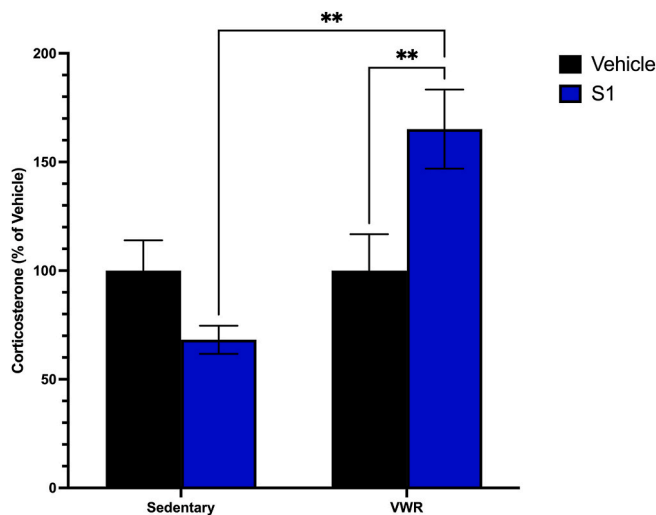


Fig. 7. Effect of VWR on Corticosterone Levels Post-S1-treatment (18d post-treatment). Rats were injected ICM with vehicle (0.9% saline) or S1 (1.0 μ g). Data are reported as a % change from sedentary vehicle animals. N = 5 per sedentary group; N = 10 per VWR group. Main effects and post-hoc comparisons are depicted in the figure, **p < 0.01.

et al., 2013, 2016a, 2017). Although beyond the scope of the current study, these results suggest that inappropriate neuroinflammation contributes to PASC symptoms by altering specific neurocircuitry (Niemczak et al., 2025; Ran et al., 2026).

4.4. PASC and hypocortisolism

Glucocorticoid dysregulation may also contribute to the behavioral symptoms of PASC (Peter et al., 2025; Troscher et al., 2024; Zheng et al., 2024). Recent reports have documented decreased serum glucocorticoid levels (e.g., cortisol) in patients with PASC, prompting growing interest in adrenal disruption as a contributing factor (Klein et al., 2023; Marques et al., 2023; Porntharukchareon et al., 2024). Hypocortisolism can cause fatigue, and in Addison's disease or adrenal insufficiency, fatigue can be debilitating (Garcia-Gonzalez et al., 2024). Furthermore, similar to PASC, patients with myalgia encephalomyelitis/chronic fatigue syndrome (ME/CFS) who experience extreme fatigue after physical activity that was not previously challenging (Berardi et al., 2026) exhibit hypothalamic-pituitary-adrenal (HPA) dysfunction (Woo et al., 2026; Zhang and Wang, 2024).

Animal experiments have established that SARS-CoV-2 infection or antigen injection reduces serum corticosterone/cortisol concentrations and, more importantly, brain concentrations of these hormones. Frank et al. reported that 9 days after ICM S1, rats had lower brain corticosterone concentrations than vehicle controls (Frank et al., 2024). Fischer et al. found that cortisol levels in the cerebrospinal fluid of African green monkeys were reduced 18 weeks after SARS-CoV-2 injection (Fischer et al., 2025). Here, we show that 18 days post-S1, sedentary rats exhibit a trend toward lower hypothalamic corticosterone levels than sedentary vehicle controls.

Several clinical studies failed to detect low basal cortisol levels in blood or saliva among PASC patients, leading some to conclude that hypocortisolism plays no role in PASC (Klein et al., 2023; Peter et al., 2025; Quach et al., 2025; Troscher et al., 2024). However, it is feasible that brain cortisol, rather than blood cortisol, plays a role by sustaining neuroinflammation and reducing motivated, rewarding behavior (Yu et al., 2015). VWR activates reward neurocircuitry, and evidence suggests that D1/D3 receptors and corticosterone mediate motivation for VWR (Ebada et al., 2016).

4.5. Physical activity, neuroinflammation, and brain corticosterone

Regular physical exercise is a potent modulator of neuroinflammation across multiple contexts (Alves et al., 2025; Connolly et al., 2022; Hu et al., 2024; Xiao et al., 2021). Furthermore, VWR is sufficient to improve outcomes in models of inflammatory disease, including multiple sclerosis, rheumatoid arthritis, and stroke (Aoyagi et al., 2025; Courel-Ibanez et al., 2026; Kinoshita et al., 2026; Zhang et al., 2026).

Physical activity could affect PASC-associated neuroinflammation and symptoms in numerous ways. We previously reported that S1 reduces corticosterone levels across multiple brain regions at 7 and 9 days post-treatment (Frank et al., 2024). Here, we demonstrate that the S1-associated reduction in brain corticosterone is diminished 18 days post-S1 in sedentary rats, whereas in VWR rats, S1 elevates brain corticosterone. S1 may induce a prolonged disruption of glucocorticoid-mediated anti-inflammatory effects that is reversed by exercise. VWR may alter glucocorticoid receptor distribution in the hypothalamus, thereby facilitating greater corticosterone-mediated anti-inflammatory effects. Moreover, exercise may alter the activity or availability of 11 β -hydroxysteroid dehydrogenase type 1, an intracellular enzyme that converts inactive cortisone to active cortisol (Seckl, 1997). This study was not designed to unravel mechanisms. Future work is needed to clarify the contribution of corticosterone in the brain to the regulation of neuroinflammation, physical activity, and diurnal activity cycles.

4.6. Limitations

This study has limitations. We tested only male rats, and evidence suggests that the neural mechanisms underlying reduced home-cage activity after LPS differ between males and females (Alencar et al., 2026). Further, mechanistic interpretations and clinical extensions of our results, including exercise intolerance, dopaminergic and glucocorticoid involvement, and activity 11 β -HSD1, are speculative. Future studies are encouraged to build on our findings with direct mechanistic manipulations. Additionally, euthanasia in mice and rats (pentobarbital) can affect neuroinflammatory endpoints. Finally, while our findings have implications for PASC, the animals were never infected with SARS-CoV-2. Nonetheless, SARS-CoV-2 antigens, including S1, are present in blood, are immunogenic, and produce long-lasting changes in gene expression indicative of microglial priming and activation, as well as reduced or disrupted diurnal wheel-running behavior.

4.7. Clinical implications

Several groups are developing exercise training methods as a therapy for PASC (Cavigli et al., 2022; Jie Chee and Dalan, 2023). Rocha et al. report a protective effect of continuous physical activity among PASC patients with musculoskeletal, neurological, and respiratory impacts (Rocha et al., 2023). Centorbi et al. surveyed 1004 participants aged 40 or older and reported that those who engage in regular physical activity are better protected against PASC symptoms (Centorbi et al., 2024). Additionally, Sarmento et al. found lower levels of fatigue, chest pain, and dizziness among individuals with PASC symptoms who walked 5000 steps or more the previous day (Sarmento et al., 2025). However, a key challenge in implementing exercise therapies is overcoming exercise intolerance in PASC patients. The present evidence suggests that exercise might alleviate some symptoms of PASC by mitigating neuroinflammatory processes and may inform investigations into effective interventions designed to reduce the brain-mediated symptoms of PASC.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT to

consolidate the abstract to meet the word limit. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: M.G.F., S.F.M., M.F. reports financial support was provided by PolyBio Research Foundation Inc. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbih.2026.101331>.

Data availability

Data will be made available on request.

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